

Simple scheme to extract maximum power from series connected photovoltaic modules experiencing mismatched operating conditions

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Abstract: Solar photovoltaic (PV) arrays formed by series combination of several PV modules suffer from considerable reduction in power yield when subjected to mismatched operating conditions. To address this issue several schemes have been reported in the literature. Amongst these schemes, the distributed maximum power point tracking (DMPPT) schemes seem to be the most promising ones. However, majority of the existing DMPPT schemes employ voltage-equalisation technique, which cannot attain exact MPP for all the modules. To overcome this problem individual power measurement units are employed for each of the module but then it leads to the involvement of large number of sensors. Schemes which do not require modular level sensing have also been reported but they require to employ a very computational intensive control algorithm coupled with the need for precise communication. This study attempts to address these limitations by proposing a simpler DMPPT strategy. It requires only one conventional MPPT algorithm to extract maximum power from the overall system. The implementation of the scheme is carried out by utilising the power circuit configuration of the existing flyback-converter based standard DMPPT scheme. The viability of the scheme is ascertained through detailed simulation and experimental studies.

1 Introduction

The increasing concerns pertaining to global warming, dwindling reserve of fossil fuel and considerable reductions in price of solar PV modules have made solar photovoltaic (PV) a promising source for electricity generation [1]. In addition to this, continuous improvements in the area of power electronic converters which are required for control, conversion and conditioning of PV power is also adding momentum to the popularity of solar PV based electricity generation systems. Due to the aforementioned virtues, solar PV is now being used for numerous applications with power rating from few watts to MW levels [2]. Typical low power PV systems are generally low voltage systems which employ one or more PV modules connected in parallel. As power rating of the system increases, the PV voltage level is also increased to reduce losses in the system [3]. The voltage level is increased by connecting several PV modules in series to form a string/array. However, the performance of such systems deteriorate considerably when some or all PV modules experience different operating conditions [4]. This mismatch in operating condition can be due to several reasons namely shading from different objects including self shadow at dusk and dawn, manufacturing process error, ageing, dust/impurity layer over PV modules, orientation mismatch, etc., to name a few.

Due to these mismatched operating conditions, the current generation capacity of each module at their respective maximum power points (I_{mpp}) differs from each other. However, being connected in series, each module has to carry the same amount of current which is generally dictated by the maximum power point tracking (MPPT) algorithm, employed to maximise the power output from that string. If this MPPT algorithm sets a current (say $I_{mpstring}$) less than the I_{mpp} of a module, it generates less power than its maximum capability. On the other hand, if $I_{mpstring}$ is more than I_{mpp} of a module, even then that particular module produces less power than its maximum capability till $I_{mpstring}$ is less than the short circuit current, I_{SC} of that module. Once $I_{mpstring}$ exceeds I_{SC}

of a module, it operates in the second quadrant of its I–V characteristics curve. When a PV module operates in second quadrant, it acts as load thereby absorbing power instead of generating power. Further, if the power consumed by a module exceeds its permissible limit, module may get damaged, due to the formation of hot-spots [5, 6]. Protection against such damage can be provided by connecting a bypass diode across each PV module [5, 6]. This diode gets turned on when module voltage, during operation in its second quadrant, reaches threshold voltage of the diode. When diode gets turned on, power consumption of PV module operating in second quadrant gets restricted to a permissible value and hence it hinders formation of hot spot. Schemes having bypass diodes are very simple to implement and exhibit negligible losses when all the series connected PV modules operate in their respective first quadrant as the diodes remain turned off. However, when diode(s) gets turned on while PV modules operate under mismatched conditions, the PV string exhibits multiple local maximum power points [7]. Although multiple local MPPs exist, none of this corresponds to the sum of individual maximum power yield capability of all the PV modules. The job of MPPT algorithm, is however, to operate the string at a point wherein the power yield is maximum amongst all the local peak power points. This MPP is known as global MPP (GMPP). The conventional MPPT algorithm cannot be employed to track GMPP as it may get trapped in a local MPP instead of the GMPP. This issue can be addressed by employing more sophisticated MPPT algorithms commonly termed as global MPPT (GMPPT) algorithms which are capable of tracking the GMPP [7, 8]. However, as mentioned earlier, even operating the string at GMPP, maximum power from each of the module cannot be extracted. This is due to the fact that if the bypass diode of a module gets turned on then it will not generate any power even though it is having the capability to do so. To extract maximum power from individual series connected PV modules the concept of distributed maximum power point tracking (DMPPT) has been introduced in several schemes reported in the literature [9–26]. The DMPPT

schemes employ additional local dc–dc converters in conjunction with a central dc–dc or dc–ac converter. These local dc–dc converters attempt to operate individual PV module at their respective MPPs. This results in the formation of a single peak power point for the PV string which can be tracked by employing conventional MPPT algorithm. The aforementioned MPPT algorithm is realised by utilising the central converter which is henceforth referred to as central MPPT algorithm. The major advantages of DMPPT schemes are: (i) each PV module can be operated at their true/approximate MPP, (ii) high efficiency as the local dc–dc converters process partial power allowing most of the PV string power to be directly transmitted to the central converter, and (iii) simple implementation.

The various schemes reported in literature for DMPPT can be broadly classified as: (i) voltage equalisation based approximate MPP schemes [10–19], (ii) model based MPP schemes [20, 21], and (iii) true/exact MPP schemes [10, 22–26].

The voltage equalisation schemes operate based on the assumption that, the voltage at maximum power point, V_{mpp} for each module remains almost the same even with wide changes in operating conditions. Hence, these schemes control voltage across every series connected module at the same value irrespective of their operating conditions. The reference voltage command for each module is generated as, $V_{ref} = V_{string_ref}/n$ wherein V_{string_ref} is the reference voltage command generated from the central MPPT algorithm for the whole string and N is the number of series connected modules. These schemes are easy to implement and require only $(N+1)$ voltage sensors and one current sensor. However, as the V_{mpp} for each module is not exactly the same under mismatched operating conditions, voltage equalisation based schemes cannot extract exact maximum power from each module.

The model based MPP schemes operate by sensing values of insolation level and temperature for each module [20, 21]. The sensed values of these parameters are fed to an approximate equivalent mathematical model of the PV module which generates suitable voltage reference command for each PV module. Thus power-information of each module or of the total string is not required for functioning of these schemes. However, such scheme heavily relies on the accuracy of the function/model used to generate reference for the voltage command. Moreover such model needs to be periodically upgraded to account for the factors such as ageing.

The exact MPP schemes can operate at true MPP of each module without sensing insolation level or temperature. However, most of the schemes which come under this group need to sense power of every module to generate their reference command of V_{mpp} [22–25]. This requires $(N+1)$ number of current sensors in addition to $(N+1)$ number of voltage sensors. As a result, overall complexity and wiring requirement for the system get increased. To avoid current sensing for each module, a scheme is proposed in [26] wherein the final output of the PV string is used for MPP tracking. However, this scheme requires $(N+1)$ instances of MPPT algorithm to run simultaneously, one for each of the controllers associated with N number of local dc–dc converters and one for the central converter. In addition to increased computational burden, such schemes may lead to ambiguous result and delayed response, as the effect of simultaneous perturbations in local converters on the string output power is not decoupled. A scheme which requires a single voltage and one current sensor is proposed in [10]. It employs multistage chopper circuit to realise local dc–dc converters. In this scheme the duty ratio of $(N-1)$ local converters is kept fixed and the duty ratio of the remaining local converter is varied to maximise the string output power. After a certain interval, duty ratio the next local converter is varied to maximise power yield whereas duty ratio of remaining $(N-1)$ converters are kept fixed. This process is repeated cyclically. However, this scheme encounters several issues as follows:

- With increment in number of series connected modules the on time duty ratio of switches in the local converters increases considerably which makes the scheme impractical for strings with higher number of series connected modules.

- Several key equations of the schemes have been derived without considering parasitic elements present in the converter which might be crucial for practical implementation as the local converters are operated in open loop fashion.
- This scheme is not tolerant to shoot through fault and the effect of dead band on the performance of the scheme needs further investigation.
- This strategy cannot be applied for schemes which employ isolated local converters as the input voltage to this converters is the string voltage itself [23]. This string voltage gets changed with perturbation in duty ratio of any local converter. As the duty ratio of remaining $(N-1)$ local converters are kept the same at a given time, the voltage level at their output, that is, voltage across each PV module also changes with changes in the input voltage. Under such scenario the variation in the string output power is not solely dependent on the variation of the duty ratio of N th converter. This violates the basic principle of operation of the scheme.

To address these issues, this paper proposes a simple scheme to extract maximum power from PV string having number of series connected PV modules. This scheme is an improved version of the scheme presented in [10] and seeks to eliminate drawbacks associated with the same. The salient features of the proposed scheme are as follows:

- Reduced burden on the processor as only one conventional MPPT algorithm needs to be executed at a given time.
- Communication link among the local dc–dc converters is not required.
- Requirement of reduced number of current sensors as the scheme does not require information of individual power from the modules. It requires only $(N+1)$ voltage sensors and only one current sensor to estimate total output power of the string.
- This scheme, simultaneously, has the effectiveness of the true/exact MPP schemes and the simplicity of implementation similar to that of the voltage equalisation based schemes

The power circuit configuration employed for the scheme is presented in the next section. The proposed control algorithm is explained in Section 3. The simulated performance is presented in Section 4 to verify the efficacy of the scheme. Experimental validation of the scheme is carried out by utilising a scaled down laboratory prototype and relevant results are presented in Section 5.

2 Power circuit configuration

The power circuit configuration of Fig. 1 which is reported in [11, 23], is utilised to demonstrate the proposed DMPPT strategy. However, the proposed DMPPT strategy can be applied for any scheme wherein each of the module voltage can be controlled by a local dc to dc converter. The power circuit configuration of Fig. 1 is having three serially connected PV modules (can be extended to N number of modules) having a flyback converter for each module. Flyback converters are employed as local dc–dc converters to operate each PV module at their respective MPPs. The module in the string which is sourcing the maximum current sets its own current level to be the string current, I_{string} . The difference between the string current and the current generated by individual PV module is termed as balance current. Each of the flyback converters has to supply the required balance current to ensure that the corresponding PV module operates in MPP. A buck-boost converter is employed as the central converter to for impedance matching with the load. The net power derived from the string is dumped into a resistive load connected across the capacitor, C_6 . Thus a stand-alone or off-grid type system is employed for demonstration of the scheme. In case of a grid-connected scheme the buck-boost converter will be replaced by a dc–ac inverter and the operation of rest of the scheme will remain the same as that of an off-grid system.

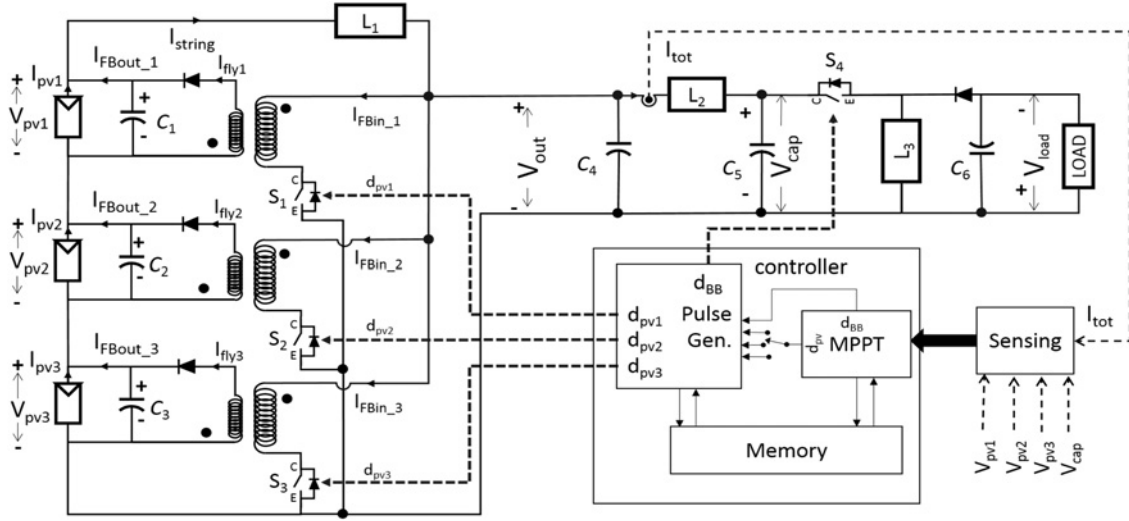


Fig. 1 Schematic circuit diagram of flyback converter based DMPPT scheme adopted for demonstration of the proposed MPPT scheme [11, 23]

The various sensed parameters required to implement the proposed DMPPT strategy are as follows: three PV module voltages (V_{pv1} , V_{pv2} , and V_{pv3}), voltage across the capacitor C_5 , V_{cap} and current through inductor L_2 , I_{tot} . The other parameters are defined as follows:

I_{pv} = PV generated currents
 I_{FBout_i} = balancing current supplied by the i th flyback converter
 I_{FBin_i} = input current to i th flyback converter
 I_{string} = string current = $\max(I_{pv1}, I_{pv2}, I_{pv3})$
 V_{out} = voltage across C_4
 V_{load} = voltage across C_6
 P_{outFB_i} = power at the output end of i th flyback converters
 $P_{tot} = I_{tot}V_{tot}$ = Power output from overall string = power input to central converter

2.1 Mathematical analysis

Under steady state conditions, the output voltage expression for the i th flyback converters operating in continuous conduction mode is

$$V_{pvi} = kV_{out} \frac{d_{pvi}}{(1 - d_{pvi})}$$

wherein k is the turns ratio of flyback transformer from output (PV side) to input side, d_{pvi} is the duty ratio of switches controlling flyback operation. Thus V_{pvi} increases with increment in d_{pvi} .

Let there be a case such that the current generation capability of module-1 is more than the other two modules. In this case the string current is,

$$I_{string} = I_{pv1} \quad (1)$$

Therefore, the balance current required from the flyback converter-1 is

$$I_{FBout_1} = I_{string} - I_{pv1} = 0 \quad (2)$$

Neglecting the voltage drops across L_1 and L_2 (which are used to filter the current ripples)

$$V_{cap} = V_{out} = V_{pv1} + V_{pv2} + V_{pv3} \quad (3)$$

The expressions for the power output from the flyback converters are

$$P_{outFB_1} = V_{pv1} \times I_{FBout_1} = 0 \quad (4)$$

$$P_{outFB_2} = V_{pv2} \times I_{FBout_2} = V_{pv2}(I_{string} - I_{pv2}) \quad (5)$$

$$P_{outFB_3} = V_{pv3} \times I_{FBout_3} = V_{pv3}(I_{string} - I_{pv3}) \quad (6)$$

Assuming the flyback converters to be lossless

$$P_{outFB_i} = P_{inFB_i} = I_{inFB_i}V_{out}$$

From (4)–(6),

$$I_{FBin_1} = 0 \quad (7)$$

$$I_{FBin_2} = V_{pv2} \frac{(I_{string} - I_{pv2})}{V_{out}} \quad (8)$$

$$I_{FBin_3} = V_{pv3} \frac{(I_{string} - I_{pv3})}{V_{out}} \quad (9)$$

From Fig. 1, considering the average current flowing through the capacitor, C_4 to be zero over a switching cycle

$$\begin{aligned} I_{tot} &= I_{string} - (I_{FBin_1} + I_{FBin_2} + I_{FBin_3}) \\ &= I_{string} - \left[0 + V_{pv2} \frac{(I_{string} - I_{pv2})}{V_{out}} + V_{pv3} \frac{(I_{string} - I_{pv3})}{V_{out}} \right], \end{aligned}$$

using (7) – (9)

$$= I_{string} \left(1 - \frac{V_{pv2}}{V_{out}} - \frac{V_{pv3}}{V_{out}} \right) + V_{pv2} \frac{I_{pv2}}{V_{out}} + V_{pv3} \frac{I_{pv3}}{V_{out}} \quad (10)$$

Using (1), (3), and (10)

$$I_{tot} = I_{string} \frac{V_{pv1}}{V_{out}} + I_{pv2} \frac{V_{pv2}}{V_{out}} + I_{pv3} \frac{V_{pv3}}{V_{out}}$$

$$\text{or, } I_{tot}V_{out} = V_{pv1}I_{pv1} + V_{pv2}I_{pv2} + V_{pv3}I_{pv3}$$

From the above equation it can be noted that the net output power from all the PV modules is obtained at the input of buck-boost converter. As there is no constraint on the voltages and currents of the PV modules, they can operate at their respective MPP points. Thus maximum power can be extracted from the system.

3 Control structure

3.1 Control philosophy

It can be noted that if a PV module is operated at a given voltage (any voltage between 0 and its open circuit voltage, V_{oc}) its output power remains constant for this particular voltage for a given operating condition. Therefore, if each of the $(N-1)$ modules is operated at a voltage fixed by their individual local converters, their total output power remains constant for a given operating condition. Hence the overall output power of the string which is given by, $P_{tot} = V_{out} I_{tot}$, is modulated by the variation of power generated in the remaining one module of the string, say i th module, whose voltage is being modulated at the current instant. Therefore, the reference voltage commands of $(N-1)$ modules are kept fixed at their respective previous updated values. The voltage reference for the i th module is adjusted such that P_{tot} and hence the output power of the i th module is ascertained to be maximum. This process is repeated for all the modules in a successive manner. Thus, at any given time the voltage reference for only one PV module is modulated/adjusted while the voltage references for the remaining PV modules in the string are kept fixed at their previous updated values. The realisation of the aforementioned control process is presented in the following section.

3.2 Realisation of the control algorithm

Based on the approach employed for adjustment of voltage references for each of the serially connected PV modules, the control algorithm can be implemented in two ways as described below.

The first approach is based on periodic search method wherein voltage references for each module are adjusted for a fixed amount of time. This fixed amount of time, T_s , is the time period of the periodic search algorithm. After every slot of T_s , the reference voltage command for the module being serviced is frozen to its updated value and the reference voltage command for the next module starts getting adjusted. The reference voltage commands for the remaining modules are kept fixed at their previous updated values. This process continues in cyclic order. As three modules are being considered here for demonstration of the proposed DMPPT strategy, the references for two modules are kept fixed for T_s , and the reference for the remaining module is adjusted to get maximum total output power, P_{tot} . The implementation of this approach is shown in Fig. 2 with the help of a simplified flowchart. At the beginning, the voltage references for the three modules, v_{m1} , v_{m2} , and v_{m3} respectively are initialised at a value, X . Although ' X ' can be any value lying between 0 to V_{oc} , however it is preferable to keep it close to the MPP of the module in question at standard test condition (STC) (as provided in the datasheet for that particular module). The integer variables j and k are initialised to zero. The variable V_r is the output of the MPPT algorithm which could be any conventional MPPT algorithm. In this paper P & O method has been used. This V_r is also initialised to a value X . The MPPT algorithm will update this V_r continuously utilising the information V_{out} and I_{tot} with an objective to maximise the string output power. Thus V_r is the reference voltage for every module being serviced for ITER_MAX number of iterations wherein ITER_MAX corresponds to time T_s . The variable k is incremented by one at the end of the time interval corresponding to ITER_MAX iterations and is reset when it reaches its maximum value n . For the very first set of iteration of j , that is, for j lying between 0 and ITER_MAX, k is zero and the reference for module 1, v_{m1} is updated with V_r . The references, v_{m2} and v_{m3} , are kept at their initialised value. Once j reaches its set limit, that is, ITER_MAX, j is reset to zero and k is increased to 1 which marks the start of the second time interval where module-2 is serviced. The previous value of v_{m2} is now loaded to V_r (for condition $k=1$ and $j=0$ as shown in Fig. 2). This step is necessary to start updating v_{m2} from its previous value otherwise v_{m2} is updated from the last value of V_r which

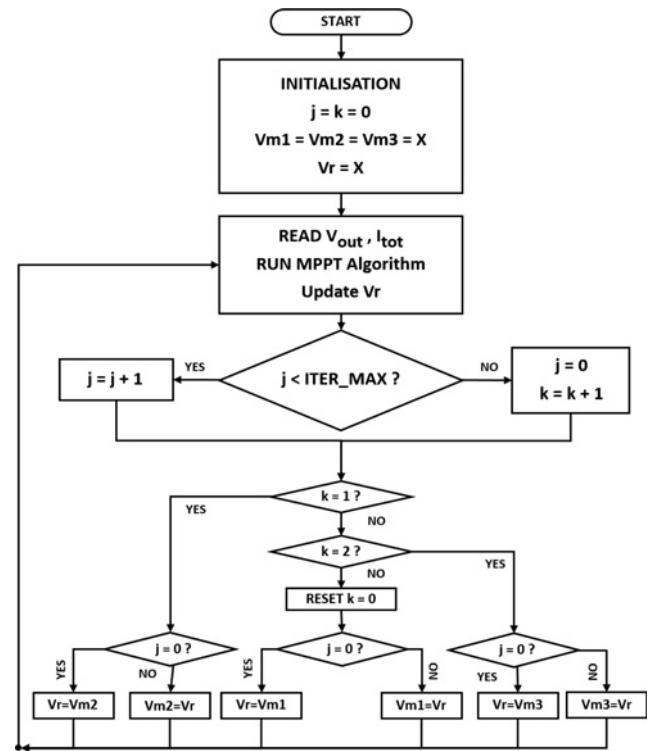


Fig. 2 Flowchart showing control structure to generate reference voltage commands for three modules being serviced at fixed interval of time (first approach)

corresponds to v_{m1} . The v_{m2} is updated till j reaches its set value ITER_MAX again. During this interval v_{m1} and v_{m3} are kept constant at their previous updated values. Once j reaches ITER_MAX again, j is reset to zero and k is increased to 2 which marks the start of the third time interval where module-3 is serviced. Similar to the previous time interval, at the beginning of this interval, previous value of v_{m3} is loaded into V_r and v_{m3} is updated till j reaches ITER_MAX. After j reaches ITER_MAX, fourth interval starts and k is reset to 0. During this time interval, v_{m1} starts getting updated while v_{m2} and v_{m3} are kept at their previous updated values and this process is repeated. Once the reference values reach their respective desired values they are kept at these values by the operation of MPPT algorithm which ensures that maximum power is drawn from the string.

In the second approach the reference voltage command for each module is adjusted till it reaches its MPP (unlike the first approach where each module is serviced for a fixed amount of time). Once MPP is reached for a given module its reference voltage command is frozen at that value and the reference voltage command for the next module is updated. The voltage references for the remaining modules are kept fixed at their previous updated values. This process continues in a cyclic order. This is realised by slightly modifying the way the variable j is updated as compared with the previous approach. The variable, j is now binary in nature which shows whether a module, being serviced, is in MPP or not. When that module is not in MPP its reference continues to get adjusted. On the other hand if MPP of that module is attained the present voltage reference is frozen and consequently the next module is being serviced. The functions of rest of the variables are same as mentioned in the first approach. The implementation of this approach is shown in Fig. 3.

From the above discussion it can be inferred that implementation of the first approach is easier compared with that of the second approach as a decision making process is required in the second approach to determine whether the MPP has attained or not. However, in case of the first approach, the waiting period (the duration of time elapsed between two successive servicing) of a module becomes considerable if the number of modules present in

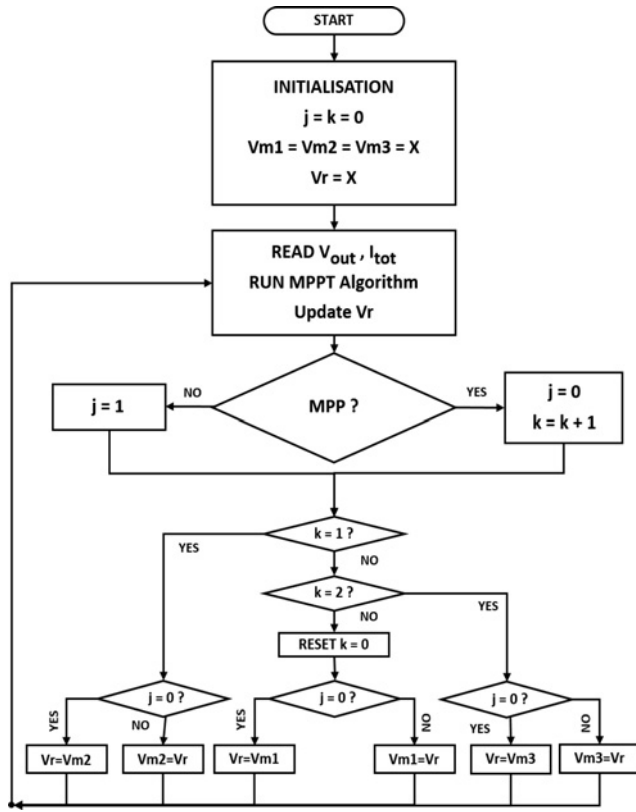


Fig. 3 Flowchart showing control structure to generate reference voltage commands for three modules with each module being serviced till it attains its MPP (second approach)

the string is large. Thus the first approach is suitable for schemes having less number of modules. In case of the second approach the time required to reach MPP for each of the modules for the first time may be considerable but then once MPP for the modules are attained or in other words steady state is reached, time required to service a module becomes less. Hence a change in operating condition in a module can be addressed quickly if the second approach is followed. The second approach is suitable for schemes having large number of serially connected modules.

The reference voltage command for the entire string is generated by adding the reference voltage commands for each of the serially connected modules and is given by

$$V_{outref} = v_{m1} + v_{m2} + v_{m3}$$

The sensed string voltage is controlled by the central dc–dc converter to follow this reference voltage command. Hence no separate MPPT algorithm is required for the central converter.

4 Simulated performance

To ascertain the viability of the proposed scheme the system shown in Fig. 1 is simulated in Matlab/Simulink platform. Three modules having similar electrical characteristics to that of ‘BP4175T’ module from BPsolar is considered for the simulation study [27].

Table 1 Electrical characteristics of BP4175T module at STC

Parameter	Value at STC
P_{mpp}	175 W
voltage at P_{mpp} , V_{mpp}	35.4 V
current at P_{mpp} , I_{mpp}	4.94 A
open circuit voltage, V_{oc}	43.6 V
short circuit current, I_{sc}	5.45 A

Table 2 Parameters considered for the simulation study

Parameter	Value
L_1 and L_2	0.1 mH
C_{pv}	100 μ F
C_{out}	500 μ F
magnetising inductance (flyback)	0.14 mH
input voltage range (flyback)	80–126 V
output voltage range (flyback)	25–40 V
max. duty ratio (flyback)	0.45
switching frequency	20 kHz

The electrical characteristics of such a module under STC is given in Table 1. The flyback converters are designed to operate in discontinuous conduction mode (DCM). The central converter is realised by a buck–boost dc–dc converter. Conventional perturb and observe (P&O) algorithm is employed to track MPP of the string. Different parameters considered for simulating the system are provided in Table 2.

Simulated performance of the scheme has been obtained for both the approaches shown in Figs. 2 and 3. Insolation levels for three modules were set at different values while maintaining a constant temperature of 25°C for each of the module for both the approaches. Relevant electrical parameters of these modules along with their set insolation levels are provided in Table 3.

4.1 Fixed time iteration approach

The fixed time interval for which each module is serviced periodically (the corresponding upper limit of j as shown in Fig. 2) has been chosen as 0.5 s. The initial voltage references for each module is set at 26 V. The voltage references generated for each module (v_{m1} , v_{m2} and v_{m3}) have been shown in Fig. 4a. From this figure it can be inferred that, during first slot of 0.5 s, v_{m1} is adjusted whereas v_{m2} and v_{m3} are kept at their initialised values, that is, 26 V. During the second slot of 0.5 s, v_{m2} is adjusted whereas v_{m1} is frozen at its previous updated value. The value of v_{m3} is also kept constant to its previous value during this period. For third slot of 0.5 s, v_{m3} is varied whereas v_{m1} and v_{m2} are kept constant at their previous updated values and this sequence of events is continued. When the references, v_{m1} , v_{m2} and v_{m3} reached their desired values (as provided in Table 3), they are maintained around these values as dictated by the P & O algorithm. The PV module voltages have been shown in Fig. 4b wherein it can be noted that they follow their respective references shown in Fig. 4a. The voltage reference command for the string and the actual string voltage has been shown in Fig. 4c wherein the string voltage is observed to follow its reference. The net output power from the string, P_{tot} has been shown in Fig. 4d. From this figure it can be inferred that the string output power is almost equal to the expected maximum power from the string as provided in Table 3.

The string current, i_{string} and currents flowing through each module have been shown in Fig. 5a. Since magnitude of the current at MPP is highest for module-3 as compared with other modules (Table 3), the string current and the current flowing through module-3 has to be equal. This can be inferred from Fig. 5a wherein i_{string} and i_{pv3} can be observed to be equal. Further, the currents flowing through all the three modules attain their prescribed values implying successful tracking of MPP of each module. The three balancing currents provided by the flyback

Table 3 Relevant details of three PV modules considered for simulation study

Module	S , kW/m ²	V_{mpp} , V	I_{mpp} , A	P_{mpp} , W
mod 1	0.8	34.8	3.97	138
mod 2	0.6	34	3.03	103
mod 3	1	35.4	4.94	175
		$\Sigma = 104.2$		$\Sigma = 416$

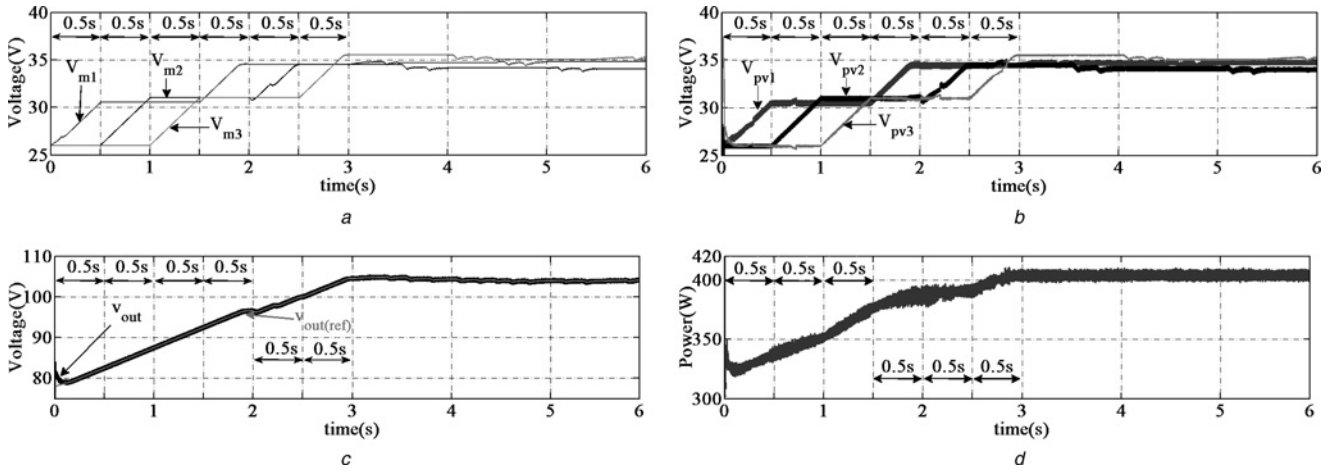


Fig. 4 Simulated response of the system (fixed time approach)

- a Reference voltage commands generated for three PV modules
b Actual voltage across three PV modules
c Reference command and actual voltage across the string
d Total power output from the string

converters and the string current, i_{string} has been shown in Fig. 5b. From this figure it can be inferred that the balancing current provided by the flyback converter connected across module-3 is zero as i_{pv3} is equal to i_{string} . Therefore this flyback converter does not process any power. The flyback converter connected to module-2 provides a balance current given by, $i_{FBout2} = i_{string} - i_{pv2} = 1.91$ A and hence the magnitude of power being processed by it is given by, $i_{FBout2} v_{pv3} = 65$ W. The flyback converter connected to module-1 provides a balance current given by, $i_{FBout1} = i_{string} - i_{pv1} = 0.97$ A and hence the magnitude of power being processed by it is given by, $i_{FBout1} v_{pv1} = 33.8$ W.

4.2 Variable time iteration/service until MPP approach

The time for which each module is serviced (see Fig. 3) is not fixed and depends upon the time duration that a given module takes to reach MPP from its current state. For the sake of fair comparison with fixed time approach, the initial voltage references for each module (v_{m1} , v_{m2} and v_{m3}) have been shown in Fig. 6a. From this figure it can be inferred that at the beginning of execution of the program, module-1 is serviced and its reference voltage v_{m1} is adjusted till it reaches its MPP. The reference voltage commands, v_{m2} and v_{m3} are kept at their initialised values, that is, 26 V while module-1 is serviced. From Fig. 6a it can be observed that at around 0.95 s the MPP of module-1 is attained. Once it is ensured that the MPP of the module-1 is reached, the controller starts servicing the module-2. While servicing module-2, v_{m1} is kept fixed at its previous updated value. The value of v_{m3} is also kept constant to its previous value during this period. The controller services module-2

till it reaches its MPP which takes about 0.88 s which can be inferred from Fig. 6a. Once it is ensured that the MPP of the module-2 is reached, the controller starts servicing the module-3. Module-3 is serviced till it reaches its MPP which takes around 0.92 s as shown in Fig. 6a. While v_{m3} is varied, v_{m1} and v_{m2} are kept constant at their previous updated values. Once MPP of module-3 is reached, the controller starts servicing module-1 again, and this process continues. The reference voltage command for the three modules, v_{m1} , v_{m2} and v_{m3} are provided in Table 3. From the Fig. 6a, it can be inferred that once the MPP of each module is reached, the duration of time elapsed between two successive servicing of a module becomes less as all the modules are already being operated at their respective MPPs. Hence any change in operating condition on a module can be addressed within a short period. The PV module voltages have been shown in Fig. 6b wherein it can be noted that they follow their respective references shown in Fig. 6a. The command for the voltage reference of the string and the actual string voltage have been shown in Fig. 6c wherein the string voltage is observed to be following its reference. The net output power from the string, P_{tot} is shown in Fig. 6d. From this figure it can be inferred that the string output power is almost equal to the expected maximum power from the string as provided in Table 3.

5 Experimental validation

Experimental validation of the proposed scheme is carried on a scaled down laboratory prototype built for the scheme shown in Fig. 1. Different components/parameters utilised for developing

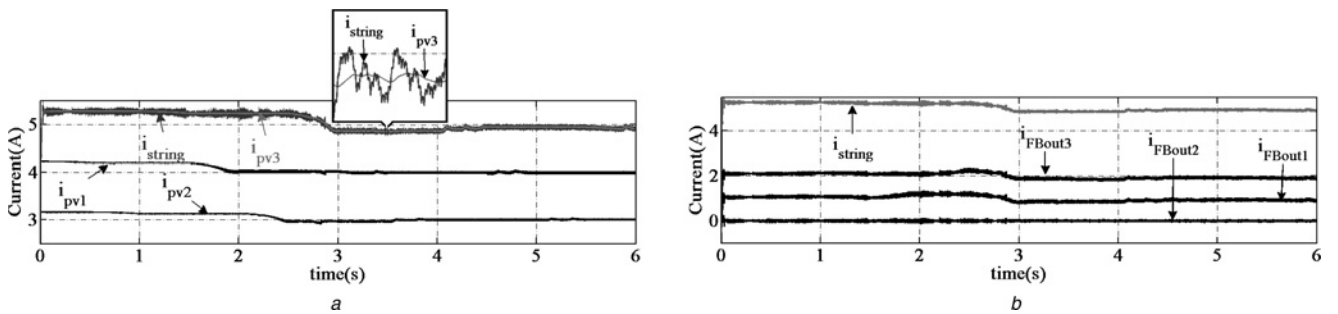


Fig. 5 Simulated response of the system (fixed time approach)

- a String current and the currents flowing through three modules
b String current and three balancing currents provided by the flyback converters

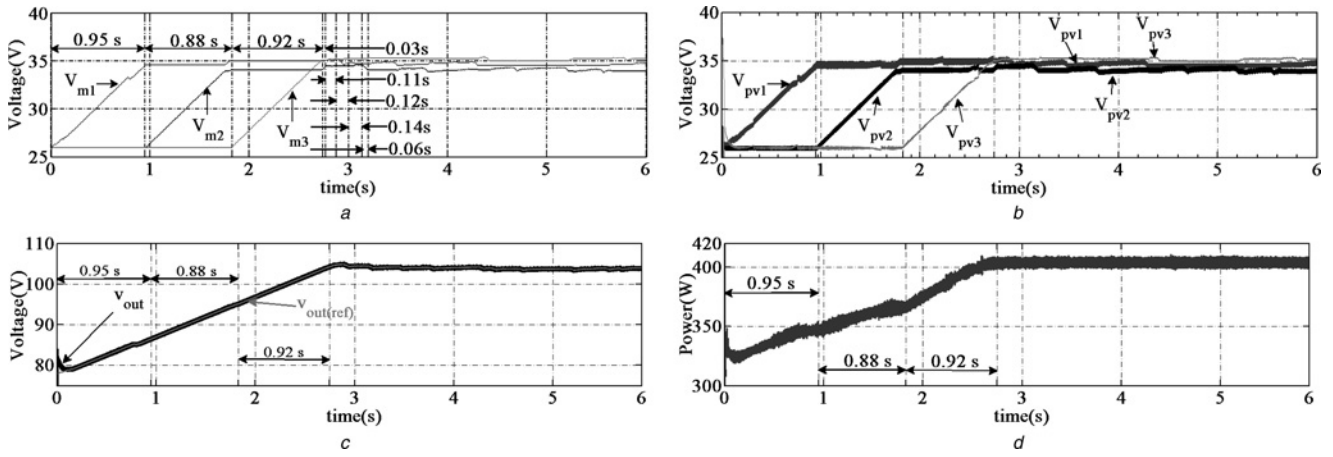


Fig. 6 Simulated response of the system (variable time approach)

- a Reference voltage commands generated for three PV modules
b Actual voltage across three PV modules
c Reference command and actual voltage across the string
d Total power output from the string

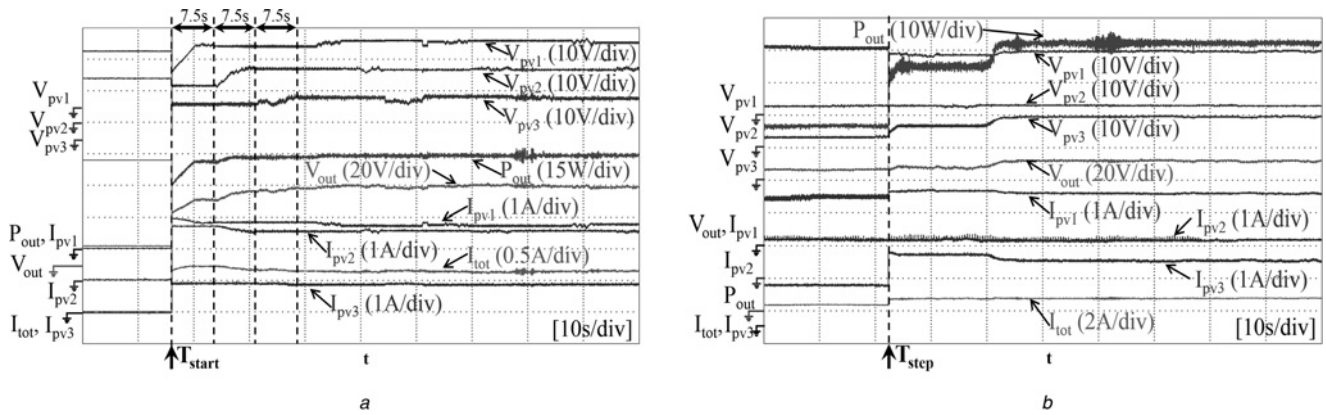


Fig. 7 Experimental results

- a Performance of the system during MPP tracking from starting of the execution of the proposed DMPPT strategy
b Performance of the system subjected to step change in insolation level in module-3

the laboratory prototype are kept same as those employed for simulation studies. The details of the semi-conductor devices employed in the scheme are as follows: (i) switches, S_1 – S_4 : IXTQ88N30P (MOSFET- 300 V, 88 A), (ii) diodes for flyback converters: FFPF30UP20STTU (200 V, 30 A), and (ii) diode for

buck-boost converter: ISL9R3060G2 (600 V, 30 A). The three PV modules are realised by employing three modules of Agilent make solar array simulator, E4360 A. These modules emulate PV characteristics of a module based on four inputs parameter, V_{oc} , V_{mpp} , I_{sc} , and I_{mpp} . The aforementioned three modules of the solar

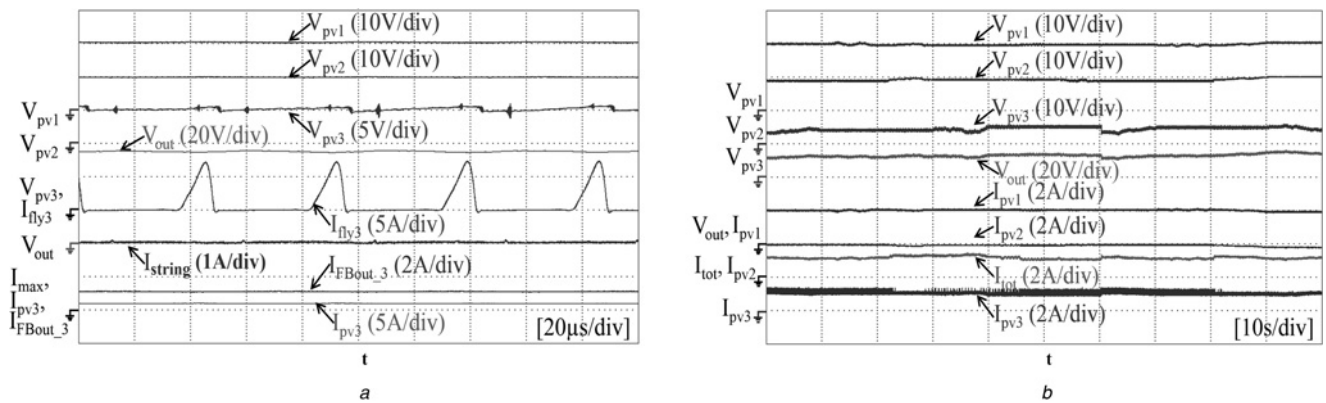


Fig. 8 Experimental results

- a Steady state performance of the system highlighting functioning of one flyback converter
b Steady state response of the system showing the output current, I_{tot} along with voltages and currents of three PV modules

Table 4 Comparison of the proposed scheme with typical DMPPT schemes sans the central converter

Scheme	n_{mpp}	n_{cs}	n_{vs}	True MPP ?	Remarks
[10]	–	1	1	yes	difficult to implement for schemes with higher N
[18]	1	1	1	no	simple to implement but cannot extract true MPP from each module
[19]	1	1	N	no	simple to implement but cannot extract true MPP from each module
[20]	N	0	N	no	needs temperature and insolation sensing
[21]	N	0	N	no	needs temperature and insolation sensing
[23]	N	$N+1$	$N+1$	yes	needs extra switch per module, PV string has to be disconnected from central converter periodically
[24]	N	$N+1$	$N+1$	yes	needs power measurement for every series connected module
[25]	–	$N+1$	$N+1$	yes	needs more current sensor, complex in control
[26]	$N+1$	1	$N+1$	yes	complex in control, needs communication between local converters
proposed	1	1	N	yes	simple to implement, reduced burden on processor, exact MPPT scheme

n_{mpp} is the number of MPP tracker required to be executed, n_{cs} is the no. of current sensor, n_{vs} is the no. of voltage sensors, and N is the number of series connected module.

array simulator are independently configurable. Hence they provide ideal platform to emulate three PV modules with different operating conditions.

The transient response of the system during tracking of MPPs of three modules has been shown in Fig. 7a. To simulate the different weather conditions, the MPP set points of the solar array simulator are set as (20.0 V, 0.8 A), (15.0 V, 1.5 A) and (15.0 V, 0.8 A) for PV1, PV2 and PV3 respectively. The open circuit voltage settings are $V_{oc1}=25$ V, $V_{oc2}=22$ V and $V_{oc3}=20$ V for PV1, PV2 and PV3 respectively. The execution of the program is started at time $t=T_{start}$. It can be seen, for $t < T_{start}$, that the electrical parameters of the three PV modules correspond to their open circuit conditions. The $V_{out}(OC)=66$ V, as expected, is approximately the summation of the three PV voltages while the current sourced by all the PV modules and the output power is practically nil. As soon as the system is started at $t=T_{start}$, the program services each module for 7.5 s to track their respective MPP points. The measured quantities upon reaching steady state are $V_{pv1}=20$ V, $I_{pv1}=0.8$ A, $V_{pv2}=17.5$ V, $I_{pv2}=1.7$ A, $V_{pv3}=15$ V, $I_{pv3}=0.9$ A. It shows that the available power at the input stage of the buck-boost converter is approximately 43 W.

Fig. 7b shows the response of the system when there is a sudden change in the insolation applied in one of the PV-modules (PV3) while the initial MPP set points are maintained to be (20.0 V, 1.5 A), (15.0 V, 1.2 A) and (15.0 V, 1.0 A) for PV1, PV2, and PV3 respectively. At the instant, T_{step} MPP voltage and current of PV3 are increased to 20 V and 2.0 A respectively and it can be observed from Fig. 7b that the system tracks the new MPP of PV3 satisfactorily.

The steady state performance of the system highlighting functioning of one flyback converter has been shown in Fig. 8a. The MPP set-points maintained are 20.0 V and 2.0 A for PV-1; 20.0 V and 2.0 A for PV-2, and 15.0 V and 1.0 A for PV-3. With these aforementioned settings it can be observed that the maximum current that can be given by any of the three PV modules is 2.0 A. Hence the string current must be 2.0 A which can be confirmed from Fig. 8a. As the current at MPP for PV-3 is 1.0 A, the flyback converter connected across it has to provide a balance current of 1.0 A to operate PV-3 at its MPP. From Fig. 8a

it can be observed that the balancing current from flyback converter-3, I_{FBout3} is 1.0 A. The current flowing through the secondary winding of the flyback transformer-3, I_{fly3} is also shown in Fig. 8a which confirms the operation of the flyback converter in DCM.

The steady state response of the system showing the output current, I_{tot} along with voltages and currents of three PV modules are provided in Fig. 8b. The MPP set-points are as follows: 20.0 V and 2.0 A for PV-1; 20.0 V and 2.0 A for PV-2, and 15.0 V and 1.0 A for PV-3. From this figure it can be noted that all the PV modules are operating at their respective MPPs. Thus it shows that the scheme is capable of tracking the true maximum power point of each module independently.

A comparison of the proposed scheme with typical DMPPT schemes is presented in Table 4. From this table it can be inferred that complexity of the control and the power circuit configuration is less in the proposed scheme. Further, the burden on the processor is least in the proposed scheme as only a conventional MPPT algorithm is required to be executed to extract maximum power from N numbers of serially connected modules. The proposed strategy can be applied for all practical purposes and as such there are no constraints imposed so far as a particular application is concerned. Hence the proposed scheme can be considered as a preferable candidate for DMPPT schemes.

6 Conclusion

This paper presents a simple DMPPT scheme capable of extracting exact maximum power from serially connected PV modules experiencing mismatched operating conditions. The main advantages of this scheme are: (i) reduced burden on the processor as only one conventional MPPT algorithm is required to be executed, (ii) elimination of communication requirement between local dc-dc converters leading to improved reliability and reduction in cost, (iii) elimination of requirement of module level power measurement leading to reduction in current sensor requirement, and (iv) simple implementation. The implementation issues and the control philosophy of the proposed scheme is discussed utilising the power circuit configuration of the existing flyback-converter based standard DMPPT scheme. The efficacy of the scheme is ascertained through detailed simulated performance. The experimental validation of the scheme is carried out on a scaled down laboratory prototype and the results are found to be satisfactory.

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